

Global Health Indicators Dashboard (2000–2024)

AnalystLab Africa Data Analytics Capstone Project

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Introduction

Health outcomes are among the clearest indicators of a country's development. Governments and international organisations rely on health data to monitor population wellbeing, evaluate healthcare systems and guide policy.

This project analyses selected health indicators from the World Bank World Development Indicators (WDI) dataset using Microsoft Power BI. The dashboard provides an interactive view of global health trends from 2000 to 2024.

Project Objective

To analyse global health indicators and identify differences in life expectancy, health expenditure, immunisation coverage and mortality across countries, regions and income groups, and to make those differences explorable by stakeholders.

Dataset Description

Source: World Bank – World Development Indicators

<https://datatopics.worldbank.org/world-development-indicators/>

The WDI dataset contains development statistics for over 200 economies across health, economy, education, environment, population, infrastructure and trade. Only health-related indicators were used in this project.

Selected indicators:

- Life expectancy at birth
- Health expenditure (% of GDP)
- Immunisation rate (DPT)
- Mortality rate

Data Cleaning and Preparation

The dataset was cleaned and transformed in Power Query before analysis:

- Imported the WDI CSV dataset into Power BI.
- Filtered the dataset to retain only the selected health indicators.
- Removed fields not required for the dashboard.
- Set appropriate data types for numerical and categorical variables.
- Built a relationship between the indicator table and the country lookup table on Country Code.
- Created measures for average life expectancy, average health expenditure, immunisation rate and average mortality rate.

Removing World Bank Aggregates

The most significant issue identified during preparation was that the WDI dataset does not separate individual countries from World Bank aggregates. Entries such as “Euro area”, “High income”, “Sub-Saharan Africa”, “Arab World” and “IDA only” appear in the same Country Name column as actual economies, with nothing marking them as different.

Left in place, these aggregates cause double- and triple-counting. Germany, for example, appears once as Germany and again within Euro area, European Union, Europe & Central Asia and High income. Every average calculated across the dataset was therefore distorted.

Real economies in the WDI country metadata carry a Region value, while aggregates have that field blank. Filtering on this reduced the country table from 264 rows to 217 individual economies. An inner join was then applied to the main indicator table so that only rows belonging to those 217 economies were retained.

This changed the headline figures across the dashboard:

- Average life expectancy: 70.67 → 71.02 years
- Average health expenditure: 6.21% → 6.25% of GDP
- Immunisation rate: 85.98% → 86.79%
- Average mortality rate: 27.72 → 26.26

Analysis Methodology

The following techniques were applied in Power BI:

- Descriptive statistics and KPI reporting
- Trend analysis over the 2000–2024 period
- Regional and income-group comparison
- Hierarchical analysis using a decomposition tree
- Interactive filtering using slicers for year, region and income group

Dashboard Overview

The dashboard contains the following elements:

- KPI cards for average life expectancy, average health expenditure, immunisation rate and average mortality rate.
- Map shading countries by income group, giving an immediate view of the global distribution.
- Combination chart comparing health expenditure (columns) with life expectancy (line) by region.
- Line chart showing immunisation rate across the study period.
- Decomposition tree allowing drill-down from region to country to income group.
- Slicers for year, region and income group.

Key Findings

- **Health expenditure varies sharply by region.** North America spends 12.95% of GDP on health, against 4.40% in South Asia (almost three times as much).
- **Life expectancy differs by more than 20 years across regions.** North America averages 79.69 years while Sub-Saharan Africa averages 59.21.
- **Income explains almost as much variation as geography.** Within Sub-Saharan Africa alone, life expectancy ranges from 73.51 years in high-income countries to 57.33 in low-income ones (a 16-year gap inside a single region).

- **Spending and outcomes do not track perfectly.** South Asia spends less on health than Sub-Saharan Africa (4.40% of GDP against 5.09%) yet averages 70.63 years of life expectancy against 59.21, a gap of over 11 years in the opposite direction to spending. This suggests that how resources are used matters alongside how much is spent.
- **Immunization coverage rose then plateaued.** Coverage climbed steadily from the early 2000s before levelling off, with a decline visible towards the end of the period.

Recommendations

1. Increase healthcare investment in regions with both low expenditure and low life expectancy, particularly Sub-Saharan Africa and South Asia.
2. Examine why some regions achieve better outcomes at similar spending levels, since efficiency of spending appears to matter as much as volume.
3. Address disparities within regions, not only between them, given the 16-year life expectancy gap between income groups inside Sub-Saharan Africa.
4. Investigate the recent decline in immunization coverage to determine whether it reflects a reporting change or a genuine drop.
5. Continue monitoring these indicators annually to support evidence based policy decisions.

Limitations

The averages in this dashboard are unweighted means across countries rather than population-weighted figures. A country with a small population therefore counts the same as one with a large population, so the figures describe the average country rather than the average person.

The dataset also contains missing values for some countries and years, particularly in earlier periods and for lower-income economies, which may affect regional averages.

Conclusion

This project demonstrates how Power BI can transform a large public dataset into an interactive tool for exploring global health outcomes. The analysis found substantial disparities in both health spending and life expectancy across regions and income groups, and identified that the relationship between the two is not straightforward.

It also highlighted the importance of understanding a dataset's structure before analyzing it. The presence of World Bank aggregates alongside individual countries was not immediately visible, and identifying and removing them was necessary before any figure in the dashboard could be relied upon.